

Tasks:

- 1. Open Kali Linux or any Linux machine.**
- 2. Execute the following commands and note their purpose:**
 - **pwd**
 - **ls**
 - **ls -la**
 - **cd**
 - **mkdir**
 - **rmdir**
 - **touch**
 - **cp**
 - **mv**
 - **cat**
 - **clear**
 - **history**
 - **whoami**
- 3. Create the following directory structure:**

CyberSecurity/

|— Notes

|— Reports

|— Tools

└— Assignments

- 4. Inside each folder, create at least 3 files using the terminal.**
- 5. Rename one file and move it to another folder.**
- 6. Copy a file from one folder to another.**
- 7. Display file contents using:**

- cat
- less

8. Find information about your system using:

- uname -a
- hostname
- ip a
- df -h
- free -h

9. Create a text file containing:

- Your Name
- Internship Name
- Today's Date

10. Use terminal commands only (avoid GUI where possible).

Assignment:

Create a report containing:

- 1. Screenshot of all created folders.**
- 2. Screenshot of terminal commands used.**
- 3. Explanation of each command.**
- 4. System information collected.**
- 5. Challenges faced and solutions.**

Without using the file manager:

- Create 10 files
- Move 5 files
- Delete 3 files
- Create 2 new folders
- Display all folder contents

Submission:

PDF or Word Document with screenshots.

Aim

To understand and practice basic Linux terminal commands, file and directory management operations, and collect system information using Kali Linux without using the graphical file manager.

Objectives

- To execute basic Linux terminal commands.
- To create and manage directories and files using terminal commands.
- To perform file operations such as copy, move, rename, and delete.
- To display file contents using terminal utilities.
- To collect system information using Linux commands.
- To perform all tasks using the command-line interface.

Basic Commands and Their Purpose

Command	Purpose
pwd	Displays the current working directory.
ls	Lists files and directories.
ls -la	Displays all files, including hidden files, with detailed information.
cd	Changes the current working directory.
mkdir	Creates a new directory.

rmdir	Removes an empty directory.
touch	Creates an empty file.
cp	Copies files or directories.
mv	Moves or renames files and directories.
cat	Displays the contents of a file.
clear	Clears the terminal screen.
history	Displays previously executed commands.
whoami	Displays the username of the current user.

1: Basic Linux Commands

```

kali@kali: ~
$ pwd
/home/kali

kali@kali: ~
$ ls
Desktop  Downloads  git-myrepo  'hash.txt'  host-10-0-2-4-discovery.txt  Music  Pictures  Public  Videos
Documents  git-demo  hash.txt  host-10-0-2-15-discovery.txt  live-hosts.gnmap  myrepo  Projects  Templates

kali@kali: ~
$ ls -la
total 244
drwxr-xr-x 24 kali kali 4096 Jun 16 01:11 .
drwxr-xr-x 13 root root 4096 May  4 02:41 ..
-rw-r--r--  1 kali kali  220 May 29 2025 .bash_logout
-rw-r--r--  1 kali kali 5551 May 29 2025 .bashrc
-rw-r--r--  1 kali kali 3526 May 29 2025 .bashrc.original
drwxr-xr-x 10 kali kali 4096 Jun  4 04:43 .BurpSuite
drwxr-xr-x 13 kali kali 4096 Jun  5 02:02 .cache
drwxr-xr-x 15 kali kali 4096 Jun  5 02:15 .config
drwxr-xr-x  2 kali kali 4096 Jun  4 03:44 Desktop
-rw-r--r--  1 kali kali  35 Jul 29 2025 .dmrc
drwxr-xr-x 13 kali kali 4096 Jul 29 2025 Documents
drwxr-xr-x  2 kali kali 4096 Jul 29 2025 Downloads
-rw-r--r--  1 kali kali 11759 May 29 2025 .face
lrwxrwxrwx  1 kali kali  5 May 29 2025 .face.icon -> .face
-rw-rw-r--  1 kali kali  71 May  4 02:51 .gitconfig
drwxrwxr-x  3 kali kali 4096 May  3 08:38 git-demo
drwxrwxr-x  2 kali kali 4096 May  4 02:52 git-myrepo
drwxr-xr-x  3 kali kali 4096 Jul 29 2025 .gnupg
-rw-rw-r--  1 kali kali  34 Dec 16 03:38 hash.txt
-rw-rw-r--  1 kali kali  34 Dec 16 03:34 'hash.txt '
-rw-r--r--  1 root root 354 Nov  6 2025 host-10-0-2-15-discovery.txt
-rw-r--r--  1 root root 353 Nov  6 2025 host-10-0-2-4-discovery.txt
-rw-r--r--  1 kali kali  10 Jul 29 2025 .ICAuthority

```

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 5
kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ rmdir TestFolder

(kali@kali)-[~]
$ touch sample.txt

(kali@kali)-[~]
$ cp sample.txt
cp: missing destination file operand after 'sample.txt'
Try 'cp --help' for more information.

(kali@kali)-[~]
$ ls
Desktop Downloads git-myrepo 'hash.txt ' host-10-0-2-4-discovery.txt Music Pictures Public Templates
Documents git-demo hash.txt host-10-0-2-15-discovery.txt live-hosts.gnmap myrepo Projects Public sample.txt Videos

(kali@kali)-[~]
$ cp sample.txt copy_sample.txt

(kali@kali)-[~]
$ ls
copy_sample.txt Documents git-demo hash.txt host-10-0-2-15-discovery.txt live-hosts.gnmap myrepo Projects Public sample.txt Videos
Desktop Downloads git-myrepo 'hash.txt ' host-10-0-2-4-discovery.txt Music Pictures Public Templates Videos

(kali@kali)-[~]
$ mv copy_sample.txt renamed_sample.txt

(kali@kali)-[~]
$ ls
Desktop git-demo 'hash.txt ' live-hosts.gnmap Pictures renamed_sample.txt Videos
Documents git-myrepo host-10-0-2-15-discovery.txt Music Projects sample.txt
Downloads hash.txt host-10-0-2-4-discovery.txt myrepo Public Templates
```

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 5
kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ cat sample.txt

(kali@kali)-[~]
$ history
1 msfconsole
2 cd Documents
3 mkdir m1 m2 m3 m4 m5
4 ls
5 cd m1
6 cat > a1
7 ls
8 history
9 pwd
10 cd ~
11 mv m3 tushar
12 mv m3 tushar3
13 mv m2 tushar
14 cd m3
15 cd Documents
16 cd m3
17 mv m3 tushar3
18 cd ~
19 mv m3 tushar3
20 ls -a
21 mv m3 tushar3
22 cd Documents
23 mkdir m1 m2 m3 m4 m5
24 mv m3 tushar3
25 ls
26 rmdir m1
27 rmdir m3
28 mkdir m1 m2 m3 m4 m5
29 rmdir m1
```

```
(kali@kali)-[~]
$ whoami
kali

(kali@kali)-[~]
$ clear
```

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 5
kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$
```

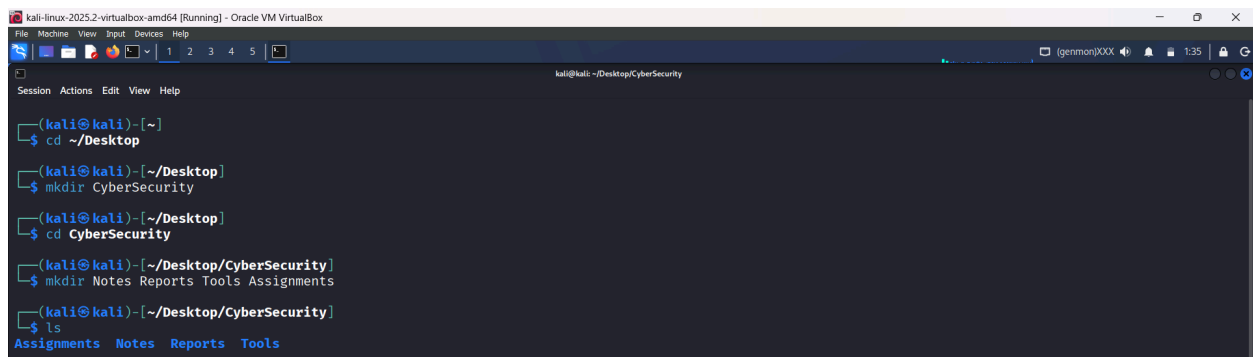
Directory Structure Creation

The following directory structure was created using terminal commands:

```
CyberSecurity/  
├── Notes  
├── Reports  
├── Tools  
└── Assignments
```

The directories were created using the mkdir command.

2: Creation of CyberSecurity Directory Structure



```
kali@kali: ~/Desktop/CyberSecurity  
$ cd ~/Desktop  
$ mkdir CyberSecurity  
$ cd CyberSecurity  
$ mkdir Notes Reports Tools Assignments  
$ ls  
Assignments  Notes  Reports  Tools
```

Creating Files in Each Directory

Three files were created inside each folder using the touch command.

Notes:

- note1.txt
- note2.txt
- note3.txt

Reports:

- report1.txt
- report2.txt
- report3.txt

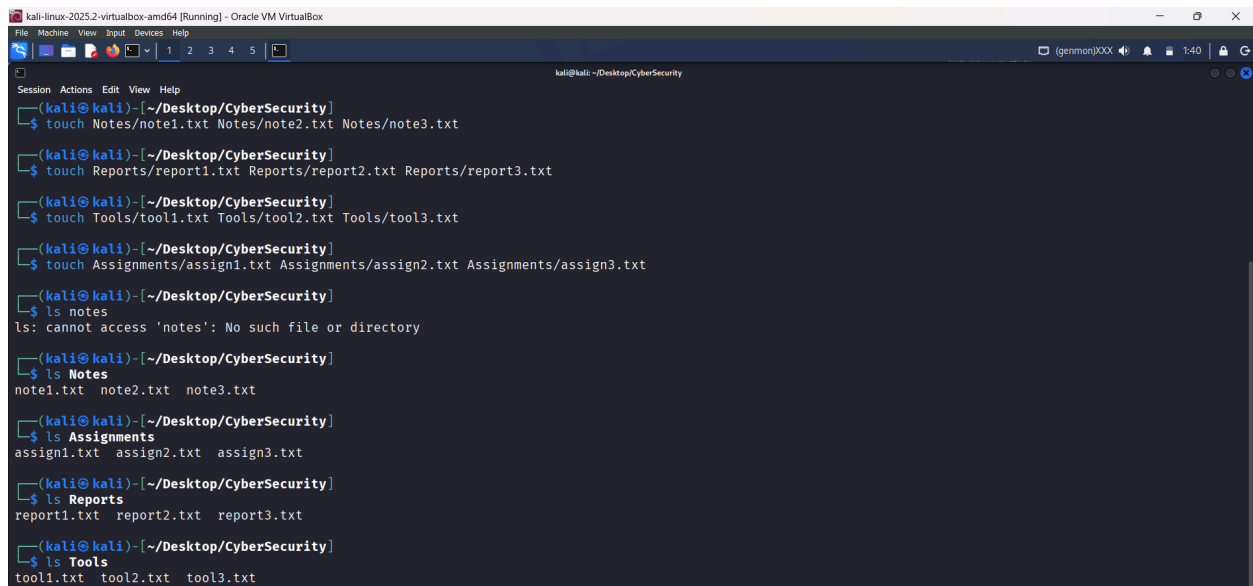
Tools:

- tool1.txt
- tool2.txt
- tool3.txt

Assignments:

- assign1.txt
- assign2.txt
- assign3.txt

3: Files Created Inside Each Folder



```
kali@kali: ~/Desktop/CyberSecurity
$ touch Notes/note1.txt Notes/note2.txt Notes/note3.txt
$ touch Reports/report1.txt Reports/report2.txt Reports/report3.txt
$ touch Tools/tool1.txt Tools/tool2.txt Tools/tool3.txt
$ touch Assignments/assign1.txt Assignments/assign2.txt Assignments/assign3.txt
$ ls notes
ls: cannot access 'notes': No such file or directory
$ ls Notes
note1.txt note2.txt note3.txt
$ ls Assignments
assign1.txt assign2.txt assign3.txt
$ ls Reports
report1.txt report2.txt report3.txt
$ ls Tools
tool1.txt tool2.txt tool3.txt
```

Renaming and Moving Files

The file note1.txt was renamed and moved from the Notes folder to the Reports folder using the mv command.

Example:

```
mv Notes/note1.txt Reports/renamed_note.txt
```

4: File Rename and Move Operation

```
(kali@kali)-[~/Desktop/CyberSecurity]
$ mv Notes/note1.txt Reports/renamed_note.txt

(kali@kali)-[~/Desktop/CyberSecurity]
$ ls Reports
renamed_note.txt  report1.txt  report2.txt  report3.txt
```

Copying Files

A file was copied from the Tools directory to the Assignments directory using the cp command.

Example:

cp Tools/tool1.txt Assignments/

5: File Copy Operation

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 5
kali@kali: ~/Desktop/CyberSecurity

(kali@kali)-[~/Desktop/CyberSecurity]
$ cp Tools/tool1.txt Assignments/

(kali@kali)-[~/Desktop/CyberSecurity]
$ ls Assignments
assign1.txt  assign2.txt  assign3.txt  tool1.txt
```

Displaying File Contents

The contents of files were displayed using the cat and less commands.

Commands Used:

cat Notes/note2.txt

less Notes/note2.txt

The cat command displays the complete contents of a file, whereas the less command allows viewing file contents page by page.

6: Displaying File Contents Using cat

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 5
/home/kali
kali@kali: ~/Desktop/CyberSecurity

(kali@kali)-[~/Desktop/CyberSecurity]
$ cp Tools/tool1.txt Assignments/

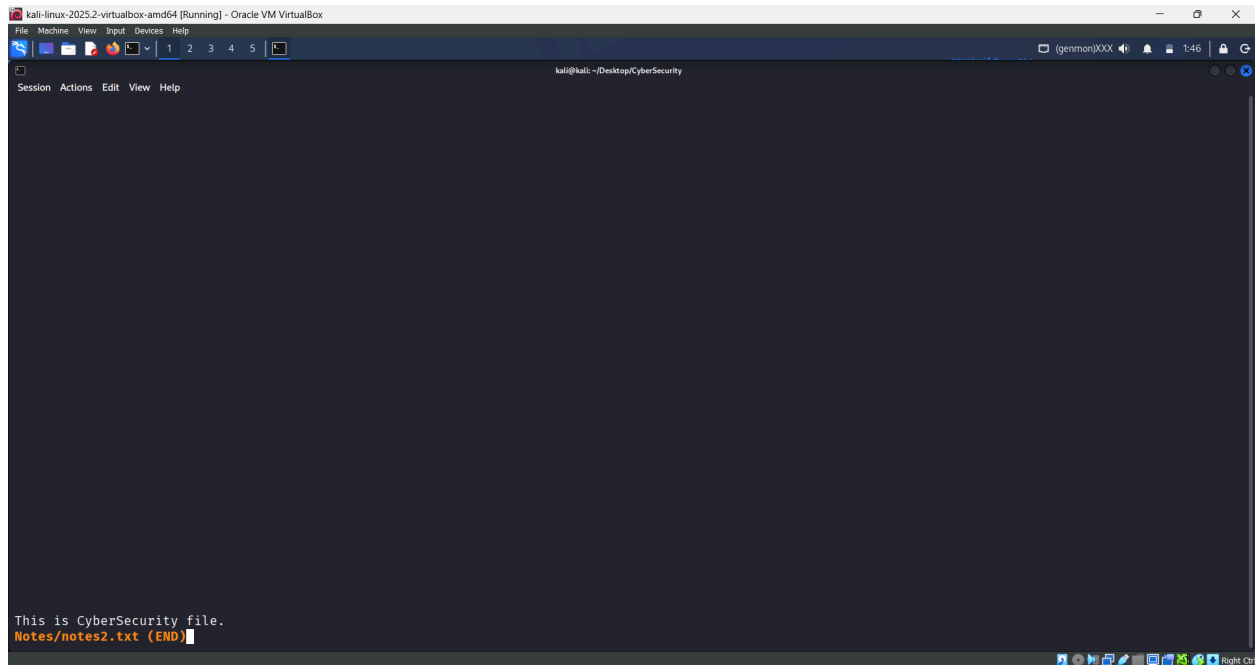
(kali@kali)-[~/Desktop/CyberSecurity]
$ ls Assignments
assign1.txt  assign2.txt  assign3.txt  tool1.txt

(kali@kali)-[~/Desktop/CyberSecurity]
$ echo "This is CyberSecurity file." >Notes/notes2.txt

(kali@kali)-[~/Desktop/CyberSecurity]
$ cat Notes/notes2.txt
This is CyberSecurity file.

(kali@kali)-[~/Desktop/CyberSecurity]
$ less Notes/notes2.txt
```


7: Displaying File Contents Using less



System Information Collection

System information was collected using the following commands:

uname -a

hostname

ip a

df -h

free -h

The purpose of each command is as follows:

- `uname -a`: Displays kernel and system information.
- `hostname`: Displays the system hostname.
- `ip a`: Displays network interface and IP address information.
- `df -h`: Displays disk usage in human-readable format.
- `free -h`: Displays memory usage information.

8: System Information

```
File Machine View Input Devices Help
kali@kali: ~/Desktop

(kali@kali)~[~/Desktop/CyberSecurity]
$ cd ..

(kali@kali)~[~/Desktop]
$ uname -a
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux

(kali@kali)~[~/Desktop]
$ hostname
kali
```

9: Network Information

```
(kali@kali)~[~/Desktop]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:d1:f8:5d brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 420sec preferred_lft 420sec
   inet6 fe80::15b:20d2:9153:b7a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

10: Disk Usage Information

```
(kali@kali)~[~/Desktop]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            881M   0 881M   0% /dev
tmpfs           197M  972K 196M   1% /run
/dev/sda1       79G   23G   52G  31% /
tmpfs           985M  4.0K 985M   1% /dev/shm
none            1.0M   0 1.0M   0% /run/credentials/systemd-journald.service
tmpfs           985M  32K 985M   1% /tmp
none            1.0M   0 1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M  116K 197M   1% /run/user/1000
```

11: Memory Information

```
(kali@kali)~[~/Desktop]
$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.9Gi          1.1Gi          241Mi          18Mi          752Mi          833Mi
Swap:          953Mi           4.0Ki          953Mi
```

Personal Information File

A text file containing personal details was created using terminal commands.

Contents:

- Name: Tushar
- Internship Name: Cyber Security Internship
- Date: 16 June 2026

The file contents were displayed using the cat command.

12: Personal Information File

```
(kali@kali)-[~/Desktop]
$ touch info.txt

(kali@kali)-[~/Desktop]
$ echo "Name: Tushargiri"> info.txt

(kali@kali)-[~/Desktop]
$ echo "Internship Name: CyberSecurity Internship">> info.txt

(kali@kali)-[~/Desktop]
$ echo "Date: 16 June 2026">> info.txt

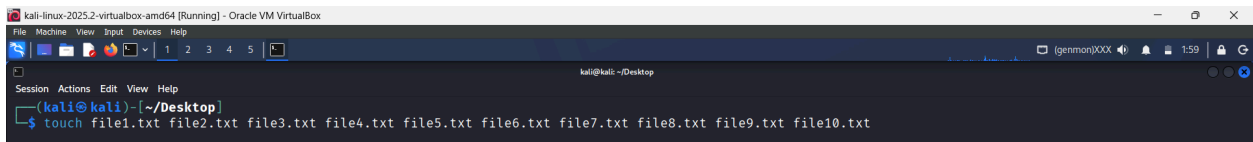
(kali@kali)-[~/Desktop]
$ cat info.txt
Name: Tushargiri
Internship Name: CyberSecurity Internship
Date: 16 June 2026
```

Without Using The File Manager:

Creating 10 Files

Ten files were created using the touch command.

13: Creation of 10 Files



```
kali@kali: ~/Desktop
$ touch file1.txt file2.txt file3.txt file4.txt file5.txt file6.txt file7.txt file8.txt file9.txt file10.txt
```

Moving 5 Files

Five files were moved to newly created folders using the mv command.

14: Moving Files



```
(kali@kali)-[~/Desktop]
$ mkdir FolderA FolderB

(kali@kali)-[~/Desktop]
$ mv file1.txt FolderA/

(kali@kali)-[~/Desktop]
$ mv file2.txt FolderA/

(kali@kali)-[~/Desktop]
$ mv file3.txt FolderA/

(kali@kali)-[~/Desktop]
$ mv file4.txt FolderB/

(kali@kali)-[~/Desktop]
$ mv file5.txt FolderB/

(kali@kali)-[~/Desktop]
$ rm file6.txt

(kali@kali)-[~/Desktop]
$ rm file7.txt

(kali@kali)-[~/Desktop]
$ rm file8.txt
```

Deleting 3 Files

Three files were deleted using the rm command.

15: Deleting Files

```
(kali@kali)-[~/Desktop]
└─$ rm file6.txt

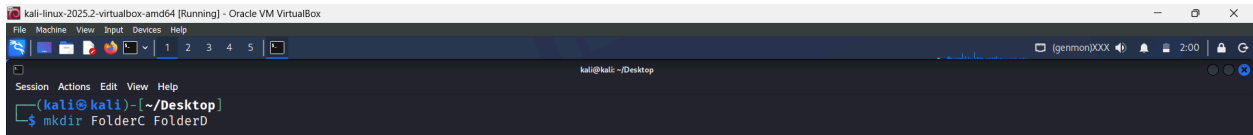
(kali@kali)-[~/Desktop]
└─$ rm file7.txt

(kali@kali)-[~/Desktop]
└─$ rm file8.txt
```

Creating 2 New Folders

Two additional folders were created using the mkdir command.

16: Creation of New Folders



```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
┌───┴───┐
└─$─┴─┐ 1 2 3 4 5 ─┘
kali@kali: ~/Desktop
Session Actions Edit View Help
└─$ mkdir FolderC FolderD
```

Displaying All Folder Contents

The ls -R command was used to recursively display the contents of all folders and subfolders.

17: Final Directory Structure

```
(kali@kali)-[~/Desktop]
└─$ ls -R
.:
CyberSecurity file10.txt file9.txt FolderA FolderB FolderC FolderD info.txt

./CyberSecurity:
Assignments Notes Reports Tools

./CyberSecurity/Assignments:
assign1.txt assign2.txt assign3.txt tool1.txt

./CyberSecurity/Notes:
note2.txt note3.txt notes2.txt

./CyberSecurity/Reports:
renamed_note.txt report1.txt report2.txt report3.txt

./CyberSecurity/Tools:
tool1.txt tool2.txt tool3.txt

./FolderA:
file1.txt file2.txt file3.txt

./FolderB:
file4.txt file5.txt

./FolderC:

./FolderD:
```

Challenges Faced and Solutions

Challenges Faced	Solutions
Permission denied error while creating a folder inside /home.	Navigated to the user's home directory using <code>cd ~</code> before creating folders.
Missing destination operand error while using <code>cp</code> .	Specified both source and destination files in the <code>cp</code> command.
Difficulty exiting with <code>less</code> command.	Pressed the <code>q</code> key to return to the terminal.
Incorrect directory navigation.	Used <code>pwd</code> and <code>ls</code> commands to verify the current location.

Conclusion

This provided hands-on experience with Linux terminal commands and file management techniques. Various operations such as creating directories, managing files, displaying file contents, collecting system information, and performing administrative tasks were successfully completed using the command-line interface without relying on the graphical file manager. The exercise improved understanding of Linux fundamentals and enhanced confidence in working within a terminal environment.